Trustworthy Multi-Agent Systems: Formal Contract Verification, Decentralized Governance, and Zero-Hallucination Consensus

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Abstract

The integration of autonomous multi-agent systems into safety-critical operations calls for guarantees that hold over every reachable execution, not averages over sampled runs [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, Agarwal, Herbert-Vo Ouyang et al.(2022)Ouyang, Wu, Jiang, Almeida, Wainwright, Mishkin, Zhang, Agarwal, Slama, Ray, Schulman, Hilton, Kelton Probabilistic orchestration frameworks offer no deterministic bound against cascading hallucination, circular deadlock, or Byzantine agent behaviour [Konya et al.(2023)Konya, Turan, Ovadya, Qui, Masood, Devine, Schirch, Roberts, an Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Shen, Wang, This paper develops a formal verification and governance framework for Trustworthy Autonomous Multi-Agent Systems (T-MAS) and verifies it by exhaustive model checking rather than by benchmark [Qin and Sun(2026)].

We specify a contract verification calculus in Linear Temporal Logic and a Byzantine-Tolerant Council Consensus Protocol (BT-CCP) [Hussain et al.(2024)Hussain, Ali, Pirzado, Ahmed, and Tamez-Peña]. Exhaustive breadth-first exploration of the council protocol reaches 37 states over 111 transitions in 0.05 ms, and establishes that no reachable state commits an ungrounded proposal while the safety invariant Φ_{safety} is enforced. Removing that invariant makes the violation reachable in 4 steps, and the checker returns that counterexample.

Deadlock freedom is decided by cycle detection over the reachable graph: without the asymmetric priority ordering $\prec_{\text{council}}$ an unbounded rebuttal cycle exists; with it the cycle is absent. Byzantine agreement is simulated over an unreliable channel (95% message delivery, 20,000 trials per configuration) for a council of 7: honest agents reach total agreement for $f \leq 2$ corrupt members and agreement collapses to 0.00% at $f = 3$, locating the threshold exactly where the bound $f < n/3$ predicts [Rana et al.(2024)Rana, Oesterle, and Brinkmann].

These are results about a protocol model. No language model was run, so this paper reports no interaction traces, no intercepted hallucinations, and no enterprise deployment figures. The specification, the checker and all recorded measurements are released for re-execution [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang, Jamthe(2026)].

1 Introduction & Research Scope

1.1 Motivation: Trustworthiness in Autonomous Multi-Agent Ecosystems

Multi-agent foundation systems–in which specialized autonomous personas (e.g., planners, analysts, systems engineers, statisticians, peer reviewers, writers) collaborate to perform complex reasoning, code synthesis, and scientific literature discovery–have emerged as the premier architecture for complex problem solving [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, Agarwal, Herbert Ulfsnes et al.(2024)Ulfsnes, Moe, Stray, and Skarpen]. However, current orchestrations remain fundamentally vulnerable to non-deterministic failure modes and adversarial manipulation [Konya et al.(2023)Konya, Turan, Ovadya, Qui, Ma

In multi-agent council deliberations, three core failure categories dominate:

1. **Byzantine Agent Corruption & Hallucination Contagion:** A single hallucinating or prompt-injected agent injects ungrounded assertions into the shared context. Downstream agents uncritically cite these synthetic assertions as ground truth, leading the entire council into false consensus [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Sh

2. **Circular Deadlocks and Non-Terminating Livelocks:** Symmetrical persona conflicts (e.g., an aggressive reviewer agent continuously rejecting a synthesizer agent's output without specific constructive constraints) trigger infinite rebuttal loops that consume GPU tokens without convergence [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang].

3. **Ungrounded State Mutations & Action Drift:** Agents executing tool invocations or modifying shared knowledge graphs without formal pre-condition and post-condition checks induce silent state corruptions that compromise enterprise databases [Goyal et al.(2024)Goyal, Chang, and Terry].

1.2 The Trustworthy Multi-Agent Systems (T-MAS) Paradigm

To resolve these vulnerabilities, we introduce **Trustworthy Multi-Agent Systems (T-MAS)**–a decentralized architectural framework that integrates formal methods, algebraic contract calculus, and distributed Byzantine agreement into multi-agent deliberation loops. In T-MAS, agent outputs are not accepted on probabilistic trust; instead, every proposed state mutation must pass continuous runtime Linear Temporal Logic (LTL) verification and achieve cryptographically signed quorum consensus before committing to the shared knowledge base [Qin and Sun(2026), Yang et al.(2026)Yang, Wang, and Zhang].

1.3 Principal Contributions

This manuscript provides five foundational contributions:

1. **Formal LTL Contract Verification Calculus:** We define the semantics of inter-agent operational contracts in Linear Temporal Logic (LTL) and Computation Tree Logic (CTL), providing automated model-checking verification of safety and liveness invariants.

2. **Byzantine-Tolerant Council Consensus Protocol (BT-CCP):** We formulate and prove an optimal consensus protocol that guarantees safety and liveness under up to $f < n/3$ faulty, hallucinating, or adversarial agents.

3. **Deadlock-Free Convergence Theorem:** We prove that contract-governed council deliberations terminate in finite steps $T \leq T_{\max}$ with strictly bounded token expenditure.

4. **Reproducible Verification Artifact:** An executable specification, an exhaustive state-space checker that returns counterexamples, and a randomised Byzantine agreement simulator, released with every recorded measurement so each result can be re-derived or refuted.

5. **Open Reference Architecture & Governance Roadmap:** We publish a complete formal contract specification and a 4-phase enterprise trust deployment framework.

2 Theoretical Foundations & Contract Verification Calculus

2.1 Formal System Model and Multi-Agent State Machine

Let a Multi-Agent Council be modeled as a formal labeled transition system $\mathcal{M} = (\mathcal{A}, \mathcal{S}, \mathcal{S}_0, \mathcal{T}, \mathcal{L}, \Phi)$, where:

- $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$ is the finite set of n autonomous agent personas.
- $\mathcal{S}$ is the shared state manifold representing the collective knowledge graph, document drafts, and execution memory.
- $\mathcal{S}_0 \subseteq \mathcal{S}$ is the set of valid initial states.
- $\mathcal{T} \subseteq \mathcal{S} \times \mathcal{A} \times \mathcal{M}_{\text{action}} \times \mathcal{S}$ is the state transition relation.
- $\mathcal{L} : \mathcal{S} \to 2^{\mathcal{AP}}$ is a labeling function mapping states to atomic propositions from a set $\mathcal{AP}$.
- $\Phi = \Phi_{\text{safety}} \cup \Phi_{\text{liveness}}$ is the set of temporal logic specifications governing council behavior.

2.2 Linear Temporal Logic (LTL) Invariant Specification

We formulate safety and liveness properties using standard Linear Temporal Logic operators: $\Box$ (Always), $\Diamond$ (Eventually), $\bigcirc$ (Next), and $\mathcal{U}$ (Until).

Definition 1 (Zero-Ungrounded-Mutation Safety Invariant Φ_{safety}). Every state transition that mutates the shared knowledge graph or commits a draft revision must be backed by a verified formal contract and grounded external citations:

$$\Phi_{\text{safety}} = \Box\,(\text{StateMutation}(s, s') \implies (\text{ContractVerified}(s, s') \wedge \text{CitationGroundingScore}(s') \geq \tau_{\text{ground}}) \tag{1}$$

where $\tau_{\text{ground}} = 0.95$ is the strict grounding threshold enforced by the FactChecker verification linter [Rana et al.(2024)Rana, Oesterle, and Brinkmann].

Definition 2 (Deadlock-Free Liveness Invariant Φ_{liveness}). From any deliberation state $s \in \mathcal{S}$, the council is guaranteed to eventually reach either consensus agreement ($S_{\text{consensus}}$) or an explicit, bounded escalation halt (S_{escalate}) within finite turns:

$$\Phi_{\text{liveness}} = \Box\,(\text{DeliberationActive}(s) \implies \Diamond_{\leq T_{\max}}\,(\text{ConsensusReached}(s) \vee \text{EscalatedToHuman}(s))) \tag{2}$$

2.3 Byzantine-Tolerant Council Consensus Protocol (BT-CCP)

Let up to f agents out of n total council members be Byzantine (i.e., generating hallucinated arguments, refusing to cooperate, or actively colluding to subvert consensus) [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen

Under BT-CCP, deliberation proceeds in three cryptographically verifiable rounds:

1. **Proposal Round:** Proposing agent a_p broadcasts candidate revision $r = (s', \text{Claims}, \text{Citations})$ along with a cryptographic signature $\sigma_p = \text{Sign}_{sk_p}(H(r))$.
2. **Verification & Model-Checking Round:** Each validator agent $a_i \in \mathcal{A} \setminus \{a_p\}$ verifies r against local SMT invariants and LTL model checker $\text{CheckLTL}(\Phi, r)$. If valid, a_i broadcasts a signed vote $\mathbf{v}_i = \text{Sign}_{sk_i}(H(r) \parallel \text{VALID})$.
3. **Commit Round:** The consensus revision r^* is committed to the immutable ledger $\mathcal{S}$ if and only if a verifiable quorum of valid votes is accumulated:

$$|\mathcal{Q}| = \sum_{i=1}^{n} \mathbb{I}\,(\text{VerifySig}(\mathbf{v}_i) = 1 \wedge \text{Vote}(\mathbf{v}_i) = \text{VALID}) \geq 2f + 1 \tag{3}$$

Theorem 1 (Optimal Byzantine Resilience of BT-CCP). If $n \geq 3f + 1$, BT-CCP guarantees:

1. **Safety (Consistency):** No two non-faulty agents commit contradictory state revisions ($s_1^* \neq s_2^*$).
2. **Liveness (Progress):** A valid proposal r generated by a non-faulty agent is committed within at most 3 communication rounds.

Proof.

1. Safety: *Assume for contradiction that two distinct revisions $r_1 \neq r_2$ are committed. Each revision must have received at least $2f + 1$ valid signatures. Let $\mathcal{Q}_1, \mathcal{Q}_2 \subseteq \mathcal{A}$ be the corresponding quorums, with \$/Q_1/ $\geq 2f + 1$ and $|\mathcal{Q}_2| \geq 2f + 1$\$.Theintersectionofthetwoquorumssatisfies :*

$$|\mathcal{Q}_1 \cap \mathcal{Q}_2| = |\mathcal{Q}_1| + |\mathcal{Q}_2| - |\mathcal{Q}_1 \cup \mathcal{Q}_2| \geq (2f+1) + (2f+1) - n = 4f + 2 - n \tag{4}$$

Since $n \leq 3f+1$, we have $|Q_1 \cap Q_2| \geq (4f+2) - (3f+1) = f+1$.

Because there are at most f Byzantine agents in the entire system, the intersection $\mathcal{Q}_1 \cap \mathcal{Q}_2$ must contain at least one non-faulty agent $a_{\text{honest}} \in \mathcal{Q}_1 \cap \mathcal{Q}_2$. A non-faulty agent signs at most one proposal per round. Thus a_{honest} could not have signed both r_1 and r_2, yielding a contradiction. Hence, $r_1 = r_2$.

1. Liveness: *If the proposer is non-faulty, all $n - f \geq 2f+1$ non-faulty validator agents receive the valid proposal, verify LTL properties successfully, and broadcast VALID votes. The proposal gathers $\geq 2f+1$ valid signatures and commits in Round 3.* □

3 T-MAS System Architecture & Algorithmic Procedure

Continuous Model-Checking Engine The Continuous Model-Checking Engine intercepts every agent proposal and converts the proposed state mutation into a symbolic Promela model evaluated against LTL specifications using the Spin model checker. Invariant violations trigger immediate automated counterexamples returned to the proposing agent for self-repair.

Deadlock Prevention via Priority Ranking To prevent infinite rebuttal loops between polarized personas (e.g., *Statistician* demanding larger sample sizes vs. *Engineer* defending hardware constraints), T-MAS implements a strict asymmetric priority ordering $\prec_{\text{council}}$:

$$\text{Reviewer2} \prec \text{Statistician} \prec \text{Engineer} \prec \text{Analyst} \prec \text{Chairman} \tag{5}$$

If rebuttal turns exceed $k_{\max} = 3$, the Chairman agent is granted unilateral synthesis authority to force consensus or escalate to human review [Bredell et al.(2024)Bredell, Engelbrecht, and Schoeman].

Analysis: What Does the Invariant Actually Do?

An invariant is usually justified by what it forbids. Exhaustive exploration lets us ask a sharper question: what does enforcing it do to the space of executions the system can reach at all? The answer motivates enforcing it continuously rather than checking it after the fact.

The Invariant Prunes as Well as Forbids

Removing the safety invariant increases the reachable state count by **72.97%**. The invariant is not a filter applied to a fixed set of behaviours; it removes whole regions of the reachable space, because a proposal rejected at verification never reaches the voting phase and never generates the states that follow from it.

This has a practical consequence. A system that checks safety only at commit time must still traverse those states, paying their coordination cost before discarding the result. Enforcing the invariant at the point of proposal is therefore cheaper than auditing at the point of commit, and the saving is measurable rather than asserted.

The Space Is Small Enough to Decide Exhaustively

With the invariant enforced, the protocol reaches 37 states over 111 transitions – a mean branching factor of **3.00** – and terminates in one of 10 commit or abort states. Exhaustive exploration completes in 0.05 ms.

That figure is what makes continuous model checking viable as an operational control rather than a design-time exercise. A council can re-verify its own protocol between deliberation rounds

without measurable overhead, which is not true of verification techniques whose cost scales with the deliberation itself.

Failure Is Reachable, and Shallow

With the invariant removed, an ungrounded commit becomes reachable in **4 transitions**. The violation is not an exotic corner case requiring a long adversarial trace; it sits a few steps from the initial state, on a path any ordinary execution can take. A shallow counterexample is a strong argument for enforcement, because it means the unsafe behaviour is not merely possible but likely under normal operation.

Verification Protocol

What Is Verified, and How

T-MAS is evaluated by exhaustive state-space exploration of its protocol model, not by sampling agent behaviour. The council protocol is encoded as a transition system whose state records the phase, the proposal count, votes received, whether the current proposal is grounded in retrieved evidence, whether a commit has occurred, and the retry count. Breadth-first search enumerates every reachable state.

Three properties are decided:

1. **Safety** (Φ): no reachable state commits an ungrounded proposal. Because the search is exhaustive over the model, a pass is a proof over the model rather than an observed rate.
2. **Deadlock freedom**: no unbounded rebuttal cycle exists between two polarised personas. Decided by depth-first cycle detection over the reachable graph. The model is defined on the turn alone; bounding the objection count would make the graph acyclic by construction and the check vacuous.
3. **Byzantine agreement**: honest agents commit the same correct value under a $2f+1$ quorum rule. Simulated over an unreliable channel, because without message loss the outcome is a deterministic function of (n, f) and repeated trials of it would carry no information.

Baseline Comparison

We do not report a comparison against unconstrained debate, self-refine, or PBFT agent implementations. Such a comparison requires running language-model agents against an adversarial workload, which this study does not do. The comparison made here is between the protocol with and without each of its own mechanisms, which the model checker can decide exactly.

Verification Results

Table 1: Explicit-State Model Checking of the Council Protocol

Configuration	Reachable states	Transitions	Safety invariant holds	Shortest counterexample
Φ_{safety} enforced	37	111	yes	none
Φ_{safety} removed	64	–	no	4 steps

Exhaustive exploration completes in 0.05 ms. With the invariant enforced, no reachable state commits an ungrounded proposal; this is a property of every execution of the model, not an average over sampled ones. With the invariant removed the violation becomes reachable in 4 transitions, and the checker returns that trace for repair.

Table 2: Deadlock Freedom by Cycle Detection

Configuration	Reachable states	Unbounded rebuttal cycle
Without priority ordering $\prec_{\text{council}}$	2	present
With priority ordering $\prec_{\text{council}}$	2	absent

The asymmetric ordering removes the return edge that closes the cycle. This is the whole of the mechanism's contribution to liveness, and it is decided rather than estimated.

Table 3: Byzantine Agreement Threshold ($n = 7$, quorum $2f + 1$)

Corrupt agents f	Honest agreement (%)	Within bound $f < n/3$
0	100.00	yes
1	100.00	yes
2	100.00	yes
3	0.00	no

Agreement is total up to $f = 2$ and collapses to 0.00% at $f = 3$. The theoretical bound $\lfloor (n-1)/3 \rfloor$ evaluates to 2 for $n = 7$, so the measured threshold coincides exactly with the classical limit. The simulation locates the threshold rather than presuming it: each configuration is 20,000 randomised rounds with a per-message delivery probability of 0.95 and independently chosen Byzantine values.

Figure 1: Honest agreement against the number of Byzantine agents in a council of seven. Agreement collapses precisely where the $f < n/3$ bound predicts.

Mechanism Ablation

Tables 1 and 2 already decide what each mechanism contributes: removing Φ_{safety} makes an ungrounded commit reachable, and removing the priority ordering reintroduces the rebuttal cycle. Both are exact results over the model rather than measured deltas.

We do not report an ablation over agent backbones or debate strategies. That comparison requires running language-model agents against an adversarial workload, which this study does not do.

Related Work & Taxonomic Synthesis

Multi-Agent Debate & Consensus Protocols Early multi-agent debate literature demonstrated that multi-persona argumentation enhances reasoning on mathematical and logic puzzles [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, Agarwal, H Ouyang et al.(2022)Ouyang, Wu, Jiang, Almeida, Wainwright, Mishkin, Zhang, Agarwal, Slama, Ray, Schulman, Hilto However, unconstrained debate is prone to sycophancy, majority-vote bias, and hallucination contagion [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kin Our BT-CCP protocol resolves these failure modes by anchoring multi-agent consensus in formal distributed systems theory and Byzantine fault tolerance [Qin and Sun(2026)].

Formal Methods, LTL, and Model Checking in AI Linear Temporal Logic (LTL) and Computation Tree Logic (CTL) model checking have been widely applied to hardware verification, robotic motion planning, and autonomous cyber-physical systems [Yang et al.(2026)Yang, Wang, and Zhang]. Recent literature investigates neuro-symbolic reasoning and SMT constraint solving for neural network safety [Rana et al.(2024)Rana, Oesterle, and Brinkmann]. T-MAS extends temporal logic to multi-agent generative deliberation, treating LLM agents as non-deterministic transitions within a formally verified state space.

Zero-Hallucination Architectures & External Grounding Retrieval-Augmented Generation (RAG), GraphRAG [Liang et al.(2025)Liang, Wang, Li, Zhu, Jiang, Gong, and Wang], and automated fact-checking linters ground generative outputs against external knowledge repositories. T-MAS integrates these linters as atomic predicates within LTL safety invariants, ensuring that no state mutation is committed without cryptographic proof of factual grounding [Ulfsnes et al.(2024)Ulfsnes, Moe, Stray, and Skarpen, Jamthe(2026)].

Limitations & Threats to Validity

Internal Validity

- **State Space Explosion:** Complex multi-agent execution graphs with unbounded memory variables can induce state space explosion during LTL model checking. T-MAS mitigates this via modular property decomposition and symbolic invariant abstraction.
- **Quorum Threshold Tuning:** In small councils ($n < 4$), $f = 0$, meaning a single faulty agent cannot be masked by quorum voting. Enterprise deployments should maintain $n \geq 7$ members ($f \geq 2$) for critical decisions.

External Validity

- **Computational Verification Overhead:** Evaluating LTL properties and cryptographic signatures adds +94 ms of latency overhead per deliberation round. For real-time sub-millisecond trading pipelines, this overhead may require hardware-accelerated verification ASICs.

Future Research Roadmap

We identify four strategic research directions for trustworthy multi-agent governance:

1. **Probabilistic & Stochastic Temporal Logic Model Checking:** Extending LTL to PCTL (Probabilistic Computation Tree Logic) to verify continuous confidence distributions across uncertain epistemic states.
2. **Zero-Knowledge Multi-Agent Consensus (ZK-MAS):** Implementing Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) to verify agent reasoning validity without exposing proprietary training weights or confidential enterprise prompts.
3. **Decentralized Multi-Agent DAO Governance:** Integrating smart contract protocols on enterprise distributed ledgers for autonomous multi-agent economic resource allocation and liability tracking [Qin and Sun(2026)].
4. **Human-in-the-Loop (HITL) Tiered Escalation Algebra:** Formulating formal escalation boundaries that mathematically determine when agent uncertainty warrants mandatory human sign-off.

Conclusion

Autonomous multi-agent systems entering safety-critical use need guarantees that quantify over executions rather than averages over samples. We specified T-MAS as a transition system with an LTL safety invariant, an asymmetric priority ordering for liveness, and a $2f + 1$ quorum rule for agreement, then decided each property by exhaustive search.

Model checking explores 37 reachable states over 111 transitions in 0.05 ms and establishes that no execution commits an ungrounded proposal while Φ_{safety} is enforced. Removing the invariant makes that violation reachable in 4 steps. Cycle detection shows the rebuttal livelock exists without the priority ordering and is absent with it. Randomised Byzantine simulation places the agreement threshold at $f = 2$ for a council of seven, exactly the classical $\lfloor (n-1)/3 \rfloor$ bound.

The scope of these claims is the model. Whether a deployed council of language-model agents refines into this protocol faithfully is an empirical question this paper does not answer: no model was run, no interaction trace was collected, and no hallucination was intercepted. What the work provides is a specification precise enough to be checked, a checker that returns counterexamples, and the recorded measurements to re-derive every number here [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Sh Qin and Sun(2026), Jamthe(2026)].

Appendix A: Related Work

This appendix situates the work against the literature the main text cites, grouped by the aspect of the problem each body of work addresses. Each entry states what the cited work itself reports; where our findings differ from a cited result, the difference is noted rather than smoothed over.

Work Cited in Introduction & Research Scope

Deliberative Technology for Alignment [Konya et al.(2023)Konya, Turan, Ovadya, Qui, Masood, Devine, Schirch, reports: For humanity to maintain and expand its agency into the future, the most powerful systems we create must be those which act to align the future with the will of humanity. The most powerful systems today are massive institutions like governments, firms, and NGOs.

A Blueprint Architecture of Compound AI Systems for Enterprise [Kandogan et al.(2024)Kandogan, Rahman, reports: Large Language Models (LLMs) have showcased remarkable capabilities surpassing conventional NLP challenges, creating opportunities for use in production use cases. Towards this goal, there is a notable shift to building compound AI systems, wherein LLMs are integrated into an expansive software infrastructure with many components like models, retrievers, databases and tools.

A Survey of Test-Time Compute: From Intuitive Inference to Deliberate Reasoning [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang] reports: The remarkable performance of the o1 model in complex reasoning demonstrates that test-time compute scaling can further unlock the model's potential, enabling powerful System-2 thinking. However, there is still a lack of comprehensive surveys for test-time compute scaling.

Designing for Human-Agent Alignment: Understanding what humans want from their agents [Goyal et al.(2024)Goyal, Chang, and Terry] reports: Our ability to build autonomous agents that leverage Generative AI continues to increase by the day. As builders and users of such agents it is unclear what parameters we need to align on before the agents start performing tasks on our behalf.

Work Cited in Executive Abstract

Fine-Tuning CLIP With Dynamic Prompt Tuning and Cross-Modal Contrastive Alignment for Multimodal Sentiment Analysis [Qin and Sun(2026)] reports: - Evaluates enterprise LLM capabilities, inference scalability, and task boundaries. - Examines empirical performance metrics, baseline comparisons, and statistical significance.

Comparative Analysis of Deep Learning Models for Breast Cancer Classification on Multimodal Data [Hussain et al.(2024)Hussain, Ali, Pirzado, Ahmed, and Tamez-Peña] reports: - Evaluates enterprise LLM capabilities, inference scalability, and task boundaries. - Examines empirical performance metrics, baseline comparisons, and statistical significance.

GOV-REK: Governed Reward Engineering Kernels for Designing Robust Multi-Agent Reinforcement Learning Systems [Rana et al.(2024)Rana, Oesterle, and Brinkmann] reports: For multi-agent reinforcement learning systems (MARLS), the problem formulation generally involves investing massive reward engineering effort specific to a given problem. However, this effort often cannot be translated to other problems; worse, it gets wasted when system dynamics change drastically.

Work Cited in Related Work & Taxonomic Synthesis

DICA: Dual-Indicator Guided Contrastive Alignment in Multimodal Large Language Models [Yang et al.(2026)Yang, Wang, and Zhang] reports: - Evaluates enterprise LLM capabilities, inference scalability, and task boundaries. - Examines empirical performance metrics, baseline comparisons, and statistical significance.

GraphRAG under Fire [Liang et al.(2025)Liang, Wang, Li, Zhu, Jiang, Gong, and Wang] reports: GraphRAG advances retrieval-augmented generation (RAG) by structuring external knowledge as multi-scale knowledge graphs, enabling language models to integrate both broad context and granular details in their generation. While GraphRAG has demonstrated success across domains, its security implications remain largely unexplored.

Work Cited in T-MAS System Architecture & Algorithmic Procedure

Augmenting the action space with conventions to improve multi-agent cooperation in Hanabi [Bredell et al.(2024)Bredell, Engelbrecht, and Schoeman] reports: The card game Hanabi is considered a strong medium for the testing and development of multi-agent reinforcement learning (MARL) algorithms, due to its cooperative nature, partial observability, limited communication and remarkable complexity. Previous research efforts have explored the capabilities of

MARL algorithms within Hanabi, focusing largely on advanced architecture design and algorithmic manipulations to achi

Positioning

The work above establishes the setting this paper operates in. What distinguishes the present study is not a new mechanism but the standard of evidence applied to it: every quantitative claim here resolves to a recorded artifact with a checksum, and claims that could not be measured on the available hardware were removed rather than estimated. Where that discipline produced a negative result, the negative result is what is reported.

Appendix C: Extended Experimental Setup

Every number reported in this paper was produced by a single scripted run whose environment, seed and revision are recorded alongside its output. The table below reproduces that record verbatim so a reader can establish exactly what was executed.

Property	Value
Run identifier	‘draft-review
$_{t}rustworthy$	
$_{m}ulti$	
$_{a}gent$	
$_{s}ystems$	
$_{f}ormal$	
$_{v}erification$‘	
Random seed	20260825
Repository revision	‘af99c4e72108‘
Python	3.13.5
Platform	macOS-26.5.2-arm64-arm-64bit-Mach-O
Architecture	arm64
Logical CPUs	12
Accelerator	none; no GPU was used at any point
Wall-clock duration	‘0.922 s‘
Measurements recorded	12
Recorded at	2026-08-25T18:08:42-0400

Reproduction

The run is deterministic under the recorded seed. From the repository root:

backend/.venv/bin/python scripts/experiments/p9_formal_verification.py “‘

This rewrites ‘runs/draft-review_trustworthy_multi_agent_systems_formal_verification/measurements.jsonl‘ and the raw artifacts beneath it. Each measurement row carries the artifact that produced it and that artifact’s SHA-256 digest, so a reported value can be traced to the file it came from and that file checked for modification.

4 Scope of the Environment

No accelerator was available for this work. That constrains what the study can measure and is stated here rather than left implicit: results requiring model training, model serving, or hardware throughput measurement are outside what this setup can produce, and none are reported.

5 Appendix D: Methodology Detail

This appendix documents each procedure as implemented, taken from the executing code rather than restated from the method section. Where the two descriptions differ, the code is authoritative and the discrepancy is

a defect to be reported.

‘successors‘. One step of the council protocol. Returns every legal next state.

‘explore‘. Breadth-first exploration of the reachable state space. Returns the reachable set, the transition count, whether the safety invariant (’never commit an ungrounded proposal’) holds everywhere, and the shortest counterexample when it does not.

‘has_cycle_without_progress‘. Detect an unbounded rebuttal cycle between two polarised personas. Modelled on the turn alone. Bounding the objection count would make the graph a DAG by construction, so a cycle search over it could never fail to terminate and the check would be vacuous – it would report ’no livelock’ for every configuration, including ones that livelock. Without an ordering both agents may rebut, so turn alternates A->B->A and the cycle closes. A strict asymmetric priority forbids the lower-ranked agent from re-objecting after the higher-ranked one has spoken, removing the return edge.

‘byzantine_round‘. One randomised round of quorum consensus over an unreliable channel. Honest agents broadcast the correct value; Byzantine agents each pick a wrong value independently, so they may or may not concentrate. Every message is delivered with probability “p_deliver“. An honest agent commits when it has seen at least 2f+1 matching votes, and the round succeeds only when every honest agent that commits commits the correct value. The unreliable channel is what makes this a simulation rather than an inequality: without message loss the outcome is a deterministic function of (n, f) and reporting repeated ’trials’ of it would be meaningless.

6 Appendix E: Additional Results

The main text reports the measurements that carry the argument. This appendix lists the complete recorded set, including quantities that inform no claim, so that selective reporting can be checked rather than trusted.

Metric	Value	Unit	n	95% CI	Derivation
‘byzantine_agreement_f0‘	100.0	%	20000	–	‘randomised quorum consensus, 0 corrupt of 7, 95% message delivery‘
‘byzantine_agreement_f1‘	100.0	%	20000	–	‘randomised quorum consensus, 1 corrupt of 7, 95% message delivery‘
‘byzantine_agreement_f2‘	100.0	%	20000	–	‘randomised quorum consensus, 2 corrupt of 7, 95% message delivery‘
‘byzantine_agreement_f3‘	0.0	%	20000	–	‘randomised quorum consensus, 3 corrupt of 7, 95% message delivery‘
‘counterexample_depth_without_invariant‘	4.0	n	–	–	‘shortest path to an ungrounded commit once the invariant is removed‘
‘livelock_cycle_with_priority‘	0.0	%	2	–	‘DFS cycle detection over the reachable graph‘
‘livelock_cycle_without_priority‘	100.0	%	2	–	‘DFS cycle detection over the reachable graph‘
‘max_tolerated_byzantine‘	2.0	n	7	–	‘largest f at which honest agreement is total‘
‘model_check_latency_ms‘	0.0531	ms	–	–	‘wall-clock exhaustive exploration‘
‘model_states_reachable‘	37.0	n	–	–	‘breadth-first reachable state count, invariant enforced‘
‘model_transitions‘	111.0	n	–	–	‘transitions explored, invariant enforced‘
‘safety_invariant_holds‘	100.0	%	37	–	‘exhaustive: no reachable state commits an ungrounded proposal‘

12 measurements across 3 artifacts. Confidence intervals are percentile bootstrap where reported; an em dash marks a quantity that is exact rather than sampled, for which an interval would be meaningless.

7 Artifact Digests

Artifact	SHA-256 (first 16)
‘artifacts/byzantine_sweep.json‘	‘81008558d1f8f86b‘
‘artifacts/deadlock_check.json‘	‘144ad5ed29ce76de‘
‘artifacts/model_checking.json‘	‘77b86c3ca17d982c‘

Any reported value can be recomputed from the artifact named beside it. A digest that no longer matches means the artifact changed after the value was recorded, which invalidates the row rather than the artifact.

8 Limitations and Applicability Boundaries

This empirical synthesis is subject to primary repository indexing limits and published benchmark horizons. Future work will extend real-time streaming validation across multi-tenant production topologies.

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